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The impact of green economy practices on firm innovation using data from the World Bank Enterprise Surveys

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Abstract

This study explores how adopting green economy practices (GEP) shapes firm innovation. A comprehensive GEP adoption index is developed to evaluate whether greater engagement in green practices enhances a firm's product and process innovation. The analysis covers ten types of GEPs, including improvements in heating and cooling, on-site clean energy generation, machinery upgrades, energy management systems, waste and recycling measures, air pollution controls, water management initiatives, fleet upgrades, lighting improvements, and other pollution control activities. Furthermore, the study investigates how different dimensions of GEPs relate to innovation outcomes. Using a firm-level dataset comprising 18,860 observations from the World Bank Enterprise Surveys (WBES), the logistic regression results indicate that firms with higher levels of GEP adoption are significantly more likely to innovate in both products and processes. These findings provide robust empirical evidence that environmental responsibility and innovation are complementary rather than competing objectives. The study offers important implications for managers seeking to integrate sustainability into innovation strategies and for policymakers promoting green and inclusive economic growth.

Keywords Green economy practices, Process innovation, Product innovation, WBES

JEL Classification G31 G38

1 Introduction

In recent years, the concept of the green economy has become central to global policy discussions as nations seek to balance economic growth with environmental sustainability. The United Nations Environment Programme (UNEP) defines an inclusive green economy as one that “results in improved human well-being and social equity while significantly reducing environmental risks and ecological scarcities.” Accordingly, a green economy is characterized by low carbon emissions, resource efficiency, and social inclusiveness. To operationalize these principles, firms increasingly adopt green economy practices (GEPs) as a strategic response to the global pursuit of sustainable economic growth. This pursuit, as defined by the Organisation for Economic Co-operation and



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Development (2011) and UNEP (2012), involves expanding economic output while preserving natural resources, minimizing environmental degradation, and enhancing social welfare and long-term productivity. At the firm level, implementing GEPs such as energy-efficient technologies, water recycling, and waste minimization enables firms to enhance resource productivity [1], strengthen competitiveness [2], and improve overall performance [3]. Moreover, the integration of GEPs aligns with national commitments to the Sustainable Development Goals (SDGs), particularly SDG 8 (Decent Work and Economic Growth) and SDG 12 (Responsible Consumption and Production). Firms that embrace GEPs also gain greater legitimacy among stakeholders [4] by demonstrating environmental responsibility and compliance with Environmental, Social, and Governance (ESG) standards. Ultimately, the adoption of GEPs should be viewed not only as an environmental mandate but as a strategic necessity for achieving long-term sustainability and competitiveness.

Innovation plays a pivotal role in enhancing firm competitiveness [5] and driving long-term growth [6]. A substantial body of literature suggests that firm innovation is shaped by a wide range of internal and external factors, including financial resources [7], corporate governance mechanisms [8], market competition [9], and institutional quality and regulatory frameworks [10]. In the current global economic environment, firms must continuously innovate to enhance efficiency, develop new products and processes, and sustain competitive advantage. Such efforts enable firms to respond proactively to market changes, regulatory pressures, and evolving consumer expectations, while transforming environmental and social challenges into opportunities for value creation [11].

Against this backdrop, the adoption of GEPs has emerged as an important driver of innovation. By integrating environmental practices into their operations and strategies, firms can address regulatory and societal demands [12] and build dynamic capabilities that support product differentiation and process improvement. Accordingly, the adoption of GEPs not only contributes to environmental objectives but may also enhance firms' innovative capacity, reinforce competitiveness, and promote sustainable economic performance. Building on this perspective, this study investigates the impact of GEP adoption on firm innovation using a World Bank Enterprise Surveys (WBES) database. A key advantage of the WBES database is that it enables researchers to measure firms' engagement in green economy practices in a detailed and standardized manner. Specifically, the WBES includes a Green Economy module, which captures multiple aspects of firms' environmental management. This module identifies ten dimensions of GEPs: (1) heating and cooling improvements; (2) climate-friendly on-site energy generation; (3) machinery upgrades; (4) energy management systems; (5) waste minimization, recycling, and waste management; (6) air pollution control measures; (7) water management measures; (8) upgrades of vehicles, vessels, or aircraft within the firm's fleet; (9) improvements in lighting systems; and (10) other pollution control measures. This detailed classification enables a more granular investigation of whether and which specific green economy practices are associated with firm-level innovation.

By empirically examining the relationship between GEPs and firm innovation, this study makes several important contributions to the literature. First, this study extends existing research by focusing on firms' adoption of specific GEPs and their direct effects on firm-level innovation. This aspect has received limited attention. In contrast, much of the prior literature has emphasized sustainability or environmental performance

outcomes [13] rather than innovation-related consequences. Furthermore, previous studies have often relied on proxy measures of environmental engagement, such as corporate social responsibility (CSR) ratings [14] or ISO 14000 certification [15], rather than on the actual implementation of green practices. Second, by examining firms' adoption across ten distinct dimensions of GEPs, this study enables a more granular assessment of whether and which practices are most effective in fostering innovative activity. Third, this study provides cross-country empirical evidence on the relationship between GEPs and firm innovation, thereby addressing a key gap in the literature, which has largely relied on single-country analyses [16, 17]. Drawing on a diverse sample of firms operating across multiple institutional and economic environments, the findings offer more generalizable insights into how GEPs influence innovation outcomes. This broader perspective enhances the understanding of the global relevance and effectiveness of GEPs in stimulating firm-level innovation.

This study is structured as follows. Section 2 presents the theoretical background and relevant literature on GEPs and firm innovation. Section 3 outlines the research design and methodology, describing the data, model, and variables. Section 4 reports the empirical results and provides a detailed discussion. Section 5 offers the conclusion.

2 Literature review

Many studies have examined the determinants of firm innovation, highlighting the roles of both internal firm characteristics and external environmental factors. Prior studies emphasize internal drivers such as financial resources [7], ownership structure [18], or corporate governance [8], which influence a firm's ability and willingness to invest in innovative activities. Beyond firm-specific factors, external conditions, including market competition [9] and institutional quality and regulatory frameworks [10], also play a critical role in shaping innovation outcomes. Given the context-dependent nature of innovation drivers, recent studies have focused on whether and how sustainability-oriented practices, such as adopting GEPs, serve as an additional channel through which firms enhance their innovation.

The Resource-Based View [19] posits that firms achieve competitive advantage by developing resources and capabilities that are valuable, rare, inimitable, and non-substitutable. In particular, physical capital resources, including plant, machinery, equipment, and production technology, constitute tangible assets that support firms' strategic activities. The adoption of GEPs typically entails targeted investments in such physical capital, including energy-efficient machinery, cleaner production technologies, pollution-control equipment, and upgrades to logistics and transportation infrastructure. These investments enhance the productivity and environmental efficiency of firms' tangible assets while simultaneously embedding firm-specific technological capabilities. As a result, the implementation of GEPs through physical capital upgrading can strengthen firms' resource bases and stimulate innovation by fostering the development of new capabilities and operational processes [20], and facilitate compliance with regulatory and societal expectations [12]. Complementing the Resource-Based View, the Porter Hypothesis [21] provides an innovation-oriented framework. It argues that well-designed environmental practices and regulations can drive innovation by encouraging firms to reduce inefficiencies, rethink production processes, and adopt cleaner technologies. From this perspective, adopting GEPs can generate innovation offsets that partially

or fully offset compliance costs [22], thereby reinforcing the positive link between the adoption of GEPs and firm innovation.

Furthermore, Institutional Theory explains the link between GEPs and firm innovation through the lens of institutional pressures and legitimacy. Firms often adopt GEPs in response to coercive pressures (environmental regulations), normative pressures (industry standards and professional norms), and mimetic pressures (the imitation of successful peers). These drivers encourage firms to redesign internal processes and develop new products. By conforming to these institutional expectations, firms enhance their legitimacy [23]; ultimately, this compliance-driven adaptation frequently catalyzes increased innovation [24].

While extant literature establishes a positive correlation between sustainability initiatives and innovation outcomes, several critical gaps persist. Primarily, prior research often relies on sustainability proxies, such as formal certifications, which reflect operational and managerial intent rather than the actual adoption of GEPs. For instance [14], demonstrated that firms with CSR engagement, as evidenced by high CSR ratings, possess superior innovative capabilities and higher new product introduction rates. Similarly [15], observed that ISO 14000 certification, denoting the implementation of environmental management systems, significantly augments both innovation inputs and outputs. While these findings suggest that voluntary environmental frameworks can catalyze innovation performance, the reliance on such certifications may overlook the nuances of actual green integration.

Second, previous studies have primarily assessed the impact of green practice implementation indirectly through a regulatory lens, leading to inconsistent empirical evidence. For example [25], utilized a quasi-natural experiment involving China's National Specially Monitored Firms program to demonstrate that heightened environmental supervision stimulates innovation, particularly within state-owned enterprises, exporters, and large-scale firms. In contrast [26], found that China's Carbon Intensity Constraint Policy actually suppressed corporate innovation. This inconsistency in prior findings suggests that, although environmental oversight can act as a catalyst for technological progress, rigid regulatory mandates may weaken firms' incentives to innovate by imposing substantial resource constraints.

Third, a significant body of research focuses on domain-specific practices and their respective innovation implications, particularly within energy management, emissions control, and sustainable logistics. Regarding energy efficiency [27], found that firms adopting climate-friendly management practices engage in more intensive climate-related research and development, which facilitates long-term improvements in energy performance. This finding is complemented by the work of [28], who utilized factor analysis and logistic regression on a sample of Danish manufacturing firms. They argue that product innovation and energy efficiency are endogenously determined, suggesting that firms may face strategic trade-offs when simultaneously optimizing these objectives.

Closely related to our study, recent empirical evidence examines the nexus between the adoption of green-oriented practices and firm innovation, suggesting that such integration yields significant positive synergies and innovation-related gains. For instance [29], employed partial least squares structural equation modeling to analyze survey data from 3,184 firms. The findings demonstrated that pro-environmental practices are positively associated with green process, product, and organizational innovation. Similarly

[30], argued that adopting green practices supports firm innovation, particularly when they foster the development of specialized sustainability-oriented capabilities. This empirical evidence supports the perspective that sustainability serves as a transformative catalyst, driving the emergence of novel products and processes that challenge conventional industrial paradigms [31].

Based on existing theories and empirical evidence, we propose the following hypothesis:

H1: Firms with greater adoption of GEPs are more likely to engage in innovation activities.

3 The model and sample

3.1 The sample

Our dataset is drawn from the WBES, which provides firm-level data from a representative sample of manufacturing and service firms across more than 150 countries. The standardized survey, initiated in 2006, is designed to capture a wide range of firm characteristics and indicators of the business environment. In this study, we utilize the Green Economy module, which collects data on firms' perceptions and practices related to the green economy in selected countries between 2018 and 2023. The original dataset covers 46 countries; however, four countries, namely Timor-Leste, Indonesia, Peru, and Bangladesh, were excluded due to non-standardized questions in this module. One limitation of the WBES is the absence of a consistent panel structure across countries and survey waves, which limits longitudinal analysis and the use of panel-specific econometric techniques. Nevertheless, the WBES has been widely employed in recent corporate finance and innovation studies [32–34]. After excluding missing observations, the final sample comprises 18,860 firm-level observations from 42 countries over the 2018–2023 period.

3.2 Model description

Model 1 aims to investigate how the adoption of GEP affects product innovation:

$$\text{PRODUCT}_{i,j} = \beta_0 + \beta_1 \text{GEP}_{i,j} + \beta_2 \text{CFL}_{i,j} + \beta_3 \text{DEBT}_{i,j} + \beta_4 \text{SIZE}_{i,j} + \beta_5 \text{FO}_{i,j} + \beta_6 \text{LARGE}_{i,j} + \beta_7 \text{AGE}_{i,j} + \beta_8 \text{SKILL}_{i,j} + \epsilon_i \quad (1)$$

Model 2 aims to investigate how the adoption of GEP affects process innovation:

$$\text{PROCESS}_{i,j} = \beta_0 + \beta_1 \text{GEP}_{i,j} + \beta_2 \text{CFL}_{i,j} + \beta_3 \text{DEBT}_{i,j} + \beta_4 \text{SIZE}_{i,j} + \beta_5 \text{FO}_{i,j} + \beta_6 \text{LARGE}_{i,j} + \beta_7 \text{AGE}_{i,j} + \beta_8 \text{SKILL}_{i,j} + \epsilon_i \quad (2)$$

3.2.1 Dependent variables

Firm innovation is proxied by two binary indicators: product innovation (PRODUCT) and process innovation (PROCESS). PRODUCT is derived from the survey question asking whether the firm introduced a new product or service. PROCESS captures whether the firm introduced a new or significantly improved process.

3.2.2 Independent variable

We construct a GEP index as in [35] to measure the extent of firms' engagement in GEPs. The index is based on responses to a series of questions in the Green Economy module of the WBES that ask whether the firm has adopted specific measures to reduce its environmental impact. Ten measurements are requested, namely: (1) Heating and cooling

improvements; (2) Climate-friendly energy generation on site; (3) Machinery upgrades; (4) Energy management systems; (5) Waste minimization, recycling, and waste management; (6) Air pollution control measures; (7) Water management measures; (8) Upgrades of vehicles, vessels, or aircraft in the fleet; (9) Improvement of lighting systems; and (10) Other pollution control measures. Each adopted practice is assigned a score of 1, and the index is calculated as the sum of the scores. The maximum possible score is ten, indicating that the firm has implemented all ten measures.

3.2.3 Control variables

The other firm-level control variables include cash flow (CFL), debt (DEBT), and firm size (SIZE). We also account for ownership structure through foreign ownership (FO) and the presence of large shareholders (LARGE). Additionally, we control for firm age (AGE) and human capital, proxied by the proportion of skilled workers (SKILL). Table 1 provides detailed descriptions of all variables and their corresponding survey question codes.

3.3 Methodology

We employ logistic regression to examine how the adoption of GEPs influences firm innovation. Logistic regression is an appropriate method when the dependent variable is binary. This model estimates the probability of a firm engaging in innovation as a function of a set of explanatory variables. The predicted values from the logistic regression range between 0 and 1, representing the estimated likelihood of the outcome. In this study, the dependent variable is a binary indicator equal to 1 if the firm engages in innovation activities and 0 otherwise. While the raw logistic coefficients represent changes in the log-odds, they are often exponentiated and interpreted as odds ratios: a value greater than 1 indicates that an increase in the independent variable (adopting more GEPs) is associated with higher odds of the firm engaging in innovation. In contrast, a value less than 1 suggests a negative association.

We first estimate a baseline model using the composite GEP index to capture the overall intensity of GEP adoption. We then estimate separate regressions for each GEP dimension to examine heterogeneity in the relationship between specific green practices and firm innovation outcomes. Estimating each dimension separately allows us to assess the association between a given practice and innovation without imposing a joint structure across potentially highly correlated practices. This approach mitigates concerns about multicollinearity and facilitates clearer interpretation of the effects of individual

Table 2 Statistical description

Variables	Observations	Mean	Standard deviation	Minimum	Maximum
PRODUCT	18,860	0.1745	0.3796	0	1
PROCESS	18,860	0.0978	0.2971	0	1
GEP	18,860	3.6257	3.1189	0	10
CFL	18,860	0.9078	0.2892	0	1
DEBT	18,860	0.2548	0.4357	0	1
SIZE	18,860	3.5823	1.3530	0	9.3398
FO	18,860	4.2907	18.7609	0	100
LARGE	18,860	82.7647	24.8507	1	100
AGE	18,860	2.857436	0.7307	0	5.3083
SKILL	18,860	0.4546	0.2976	0	1

Table 1 Variable description. *Source:* WBES database

Variable name	Abbreviation	Measurement	Survey item
<i>Firm innovation</i>			
Product innovation	PRODUCT	The variable takes a value of one if the firm reports introducing new or services.	h1
Process innovation	PROCESS	The variable takes a value of one if the firm reports introducing new or significantly improved process	h5
<i>GEPs</i>			
Heating and cooling improvements	GEP1	The variable takes a value of one if the firm reports implementing this measurement	BMGc23a
Climate-friendly energy generation on-site	GEP2	The variable takes a value of one if the firm reports implementing this measurement	BMGc23b
Machinery upgrades	GEP3	The variable takes a value of one if the firm reports implementing this measurement	BMGc23c
Energy management systems	GEP4	The variable takes a value of one if the firm reports implementing this measurement	BMGc23d
Waste minimization, recycling, and waste management	GEP5	The variable takes a value of one if the firm reports implementing this measurement	BMGc23e
Air pollution control measures	GEP6	The variable takes a value of one if the firm reports implementing this measurement	BMGc23f
Water management measures	GEP7	The variable takes a value of one if the firm reports implementing this measurement	BMGc23g
Upgrades of vehicles, vessels, or aircraft in the fleet	GEP8	The variable takes a value of one if the firm reports implementing this measurement	BMGc23h
Improvement of lighting systems	GEP9	The variable takes a value of one if the firm reports implementing this measurement	BMGc23i
Other pollution control measures	GEP10	The variable takes a value of one if the firm reports implementing this measurement	BMGc23j
Green Economy Practices Index	GEP	$GEP = (GEP1 + GEP2 + GEP3 + GEP4 + GEP5 + GEP6 + GEP7 + GEP8 + GEP9 + GEP10)$	
<i>Control variables</i>			
Cash flow	CFL	The variable takes a value of one if the firm reports having checking and/or saving accounts.	k6
Firm debt	DEBT	The variable takes a value of one if the firm reports having a line of credit or a loan from financial institutions	k8
Firm size	SIZE	Firm size is the natural logarithm of number of full-time workers	l1
Foreign ownership	FO	Ownership by foreign shareholders	b2b
Ownership by large shareholders	LARGE	Ownership by large owners	b3
Firm age	AGE	Firm age is calculated as the natural logarithm of the number of years elapsed from the firm's founding to the year the survey was conducted.	a14y; b5
Human capital	SKILL	The ratio of skilled workers to total employment, where total employment is defined as the sum of skilled, semi-skilled, and unskilled workers.	l4a1; l4a; l4a2; l4b

practices. All specifications incorporate country, year, and industry fixed effects to control for unobserved heterogeneity and common shocks. In all regressions, statistical inference is based on robust standard errors clustered by country. Additionally, as a robustness check, we re-estimate the models after excluding firms from India, which accounts for 39.69% of the sample, to ensure the overrepresentation of a single country does not drive the results.

4 Results and discussion

4.1 Statistical description

Table 2 presents the descriptive statistics for all variables. On average, 17.45% of firms reported introducing a product innovation, while 9.78% engaged in process innovation. The GEP index ranges from 0 to 10, indicating a wide range of adoption, from firms with no green engagement to those implementing all 10 practices. The mean value of CFL is 0.9078, indicating that 90.78% of firms report having a checking and/or savings account. In terms of DEBT, 25.48% of firms report having a loan from a financial institution or an active line of credit. FO has a mean of 4.29%, while LARGE averages 82.76%. Finally, SKILL has a mean of 0.4546, suggesting that 45.46% of the workforce consists of high-skilled workers.

Table 3 presents the distribution of observations by country. The dataset comprises 18,860 observations across 42 countries. India accounts for the largest share, with 7,486 observations (39.69%), followed by Egypt with 1,491 observations (7.91%). In contrast, the smallest shares are observed for Kosovo (47 observations; 0.25%) and Montenegro (55 observations; 0.29%).

Table 4 shows the distribution of the sample by GEP score. A significant proportion of firms (23.54%) have not adopted any GEPs (score = 0). Nearly half of the sample (46.83%) adopted between one and five practices, indicating partial engagement rather than comprehensive implementation. In contrast, only 1.26% of firms adopted all ten practices, demonstrating that full-scale green transformation remains limited across the surveyed countries.

Table 5 shows that the adoption rates of GEPs vary significantly across firms, ranging from 7.99% to 53.68%. The most widely adopted practices are those that offer direct cost

Table 3 Data distribution by country

Order	Country	Observations	Percentage	Order	Country	Observations	Percentage
1	Jordan	131	0.69	22	Mongolia	114	0.60
2	West Bank and Gaza	103	0.55	23	Estonia	79	0.42
3	Morocco	195	1.03	24	Kosovo	47	0.25
4	Albania	107	0.57	25	Czech	234	1.24
5	Belarus	304	1.61	26	Hungary	443	2.35
6	Georgia	127	0.67	27	Latvia	86	0.46
7	Tajikistan	66	0.35	28	Lithuania	102	0.54
8	Turkey	803	4.26	29	Slovak	164	0.87
9	Ukraine	740	3.92	30	Slovenia	130	0.69
10	Uzbekistan	628	3.33	31	Bulgaria	306	1.62
11	Russia	611	3.24	32	Croatia	144	0.76
12	Poland	567	3.01	33	Montenegro	54	0.29
13	Romania	334	1.77	34	Egypt	1,491	7.91
14	Serbia	82	0.43	35	Greece	264	1.4
15	Kazakhstan	694	3.68	36	Portugal	684	3.63
16	Moldova	65	0.34	37	Lebanon	236	1.25
17	Bosnia and Herzegovina	61	0.32	38	India	7,486	39.69
18	Azerbaijan	32	0.17	39	Tunisia	214	1.13
19	Macedonia	81	0.43	40	Cyprus	70	0.37
20	Armenia	179	0.95	41	Italy	420	2.23
21	Kyrgyz Republic	113	0.60	42	Malta	69	0.37

Table 4 Data distribution by GEPs

GEP score	Observation	Percentage
0	4,440	23.54
1	1,981	10.5
2	1,944	10.31
3	1,807	9.58
4	1,607	8.52
5	1,494	7.92
6	1,216	6.45
7	1,260	6.68
8	1,161	6.16
9	1,712	9.08
10	238	1.26
Total	18,860	100.00

Table 5 Adoption of GEPs

Index	GEPs	Adoption rate (%)
GEP1	Heating and cooling improvements	41.72
GEP2	Climate-friendly energy generation on site	7.99
GEP3	Machinery upgrades	49.95
GEP4	Energy management systems	41.15
GEP5	Waste minimization, recycling, and waste management	44.36
GEP6	Air pollution control measures	28.94
GEP7	Water management measures	33.01
GEP8	Upgrades of vehicles, vessels, or aircraft in the fleet	35.55
GEP9	Improvement of lighting systems	53.68
GEP10	Other pollution control measures	26.21

savings or efficiency benefits, such as lighting system improvements (53.68%), machinery upgrades (49.95%), waste minimization and recycling (44.36%), heating and cooling improvements (41.72%), and energy management systems (41.15%). Conversely, the least frequently adopted practice is climate-friendly energy generation on site (7.99%), suggesting that capital-intensive investments remain nascent. This low adoption rate is likely attributable to high upfront costs or a lack of internal technical capacity.

4.2 Results and discussion

4.2.1 Baseline results

Table 6 reports the odds ratios from logistic regression models examining the relationship between the GEP index and firm innovation outcomes. The estimates indicate that the GEP index is positively and significantly associated with innovation. Specifically, a one-unit increase in the GEP index, representing the adoption of one additional GEP, is associated with significantly higher odds of both product innovation (odds ratio = 1.157, $p < 0.01$) and process innovation (odds ratio = 1.194, $p < 0.01$). In percentage terms, the adoption of an additional GEP is associated with an increase of 15.7% and 19.4% in the odds of product and process innovation, respectively. Several underlying mechanisms can explain the positive relationship. The adoption of GEPs encourages firms to use energy, materials, and water more efficiently, prompting them to redesign production systems, implement cleaner technologies, and optimize resource flows to reduce waste and emissions [36, 37]. At the same time, firms may develop new eco-efficient products that consume fewer resources and generate lower environmental impacts in response to

Table 6 The GEP index and firm innovation

Variables	Product		PROCESS	
	Odds ratio	z-statistics	Odds ratio	z-statistics
GEP	1.157***	(5.880)	1.194***	(8.885)
CFL	1.264	(1.096)	1.640	(1.864)
DEBT	1.440***	(3.881)	1.946***	(4.913)
SIZE	1.077*	(2.446)	1.053	(1.602)
FO	1.000	(−0.389)	1.000	(0.290)
LARGE	0.997**	(−2.736)	0.998	(−1.561)
AGE	1.018	(0.564)	0.946	(−1.133)
SKILL	1.117	(1.174)	0.955	(−0.328)
Observations	18,860			

z-statistics based on robust standard errors clustered at the country level are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. All models include country, year, and industry fixed effects

growing regulatory requirements and market demand for sustainable goods [38]. Moreover, green product standards and certifications motivate firms to develop innovative, compliant products, thereby strengthening their competitive position [39]. These findings suggest that the adoption of GEPs should not be viewed as a compliance-driven response to environmental regulations, but rather as a strategic mechanism through which firms can strengthen their innovative capabilities.

From a theoretical perspective, the findings provide empirical support for Institutional Theory by showing that firms adopt GEPs not only in response to regulatory pressures but also to gain legitimacy among international stakeholders, customers, and industry peers. This legitimacy-seeking behavior encourages isomorphic learning, whereby standardized green practices, such as waste and energy management, facilitate organizational renewal and innovation. The results also align with the Porter Hypothesis, suggesting that environmental practices can stimulate innovation by improving efficiency and encouraging process redesign and cleaner technologies. In addition, the evidence is consistent with the Resource-Based View, indicating that investments in green physical capital, including machinery upgrades and energy-efficient systems, are closely associated with higher probabilities of product and process innovation, reflecting incremental learning and technological upgrading.

Importantly, the cumulative effect captured by the GEP index suggests significant complementarities among diverse green practices. Rather than acting in isolation, multiple environmental initiatives may mutually reinforce one another, creating synergies across production cycles, energy consumption patterns, and environmental management systems. Such internal complementarities can enhance absorptive capacity and facilitate the cross-functional diffusion of knowledge, thereby increasing the likelihood of innovation. This finding provides a robust explanation for why firms adopting a broader, integrated suite of green practices tend to achieve superior innovation outcomes compared to those implementing isolated or fragmented measures.

These results also contribute to the empirical literature by extending existing studies that have focused primarily on environmental or sustainability performance [13], or that have relied on proxy measures such as CSR ratings or environmental certifications [14, 15]. Moreover, the cross-country nature of the dataset suggests that the innovation-enhancing effects of GEP adoption are not idiosyncratic to a particular institutional or

economic context. Instead, they reflect a pervasive mechanism that enables environmental practices to foster innovation across diverse global settings.

Regarding the control variables, DEBT significantly and positively affects both product and process innovation, suggesting that external financing provides the capital necessary to fund innovative activities. Specifically, firms securing a new credit line or loan are associated with a 44.0% and 94.6% increase in the odds of product and process innovation, respectively. SIZE and LARGE have significant effects on product innovation; however, they are statistically insignificant for process innovation. Specifically, a one-unit increase in firm size is associated with a 7.7% increase in the odds of product innovation. Conversely, higher ownership concentration by large shareholders is associated with a slight decrease in the odds of product innovation. Other factors, including CFL, AGE, FO, and SKILL, do not demonstrate significant effects within this specification.

4.2.2 Several dimensions of GEPs

In this subsection, we estimate separate regressions for each GEP dimension to examine the heterogeneous associations between specific green practices and firm innovation outcomes. Tables 7 and 8 present the logistic regression results for the individual dimensions of GEPs on product and process innovation, respectively. The results reveal that all ten dimensions positively and significantly influence the likelihood of introducing both types of innovation. This uniform significance suggests that whether a practice is operational (e.g., improvements in lighting systems or machinery upgrades) or capital-intensive (e.g., climate-friendly energy generation), the adoption of green measures consistently serves as a catalyst for broader firm-level innovation.

Among these dimensions, machinery upgrades exhibit the most pronounced effect: adopting this practice increases the odds of product innovation by 2.419 and process innovation by 2.794. These results suggest that firms implementing machinery upgrades have 141.9% and 179.4% higher odds of introducing product and process innovations,

Table 7 The adoption of GEPs and product

	GEP1	GEP2	GEP3	GEP4	GEP5	GEP6	GEP7	GEP8	GEP9	GEP10
GEP	1.702*** (4.994)	1.594*** (5.649)	2.419*** (11.602)	1.718*** (4.144)	1.840*** (5.489)	1.480*** (5.373)	1.515*** (3.981)	1.700*** (6.473)	1.644*** (3.747)	1.528*** (5.771)
CFL	1.250 (1.090)	1.308 (1.369)	1.193 (0.906)	1.281 (1.230)	1.269 (1.162)	1.314 (1.354)	1.296 (1.305)	1.281 (1.267)	1.240 (1.075)	1.333 (1.406)
DEBT	1.516*** (4.360)	1.546*** (4.759)	1.438*** (3.861)	1.520*** (4.499)	1.514*** (4.712)	1.540*** (4.864)	1.539*** (4.744)	1.504*** (4.353)	1.524*** (4.371)	1.545*** (4.966)
SIZE	1.132*** (3.731)	1.146*** (3.897)	1.113** (3.086)	1.129*** (3.876)	1.135*** (3.397)	1.146*** (3.723)	1.143*** (3.974)	1.133*** (3.472)	1.144*** (3.881)	1.152*** (3.990)
FO	1.000 (0.012)	1.000 (0.131)	1.000 (−0.102)	1.000 (−0.149)	1.000 (−0.091)	1.000 (0.009)	1.000 (−0.046)	1.000 (0.283)	1.000 (0.027)	1.000 (0.111)
LARGE	0.997** (−2.789)	0.997** (−2.884)	0.997** (−2.837)	0.997** (−2.894)	0.998* (−2.572)	0.997** (−2.905)	0.997** (−2.981)	0.997** (−2.750)	0.997** (−2.756)	0.997** (−2.928)
AGE	1.030 (0.936)	1.031 (1.035)	1.036 (1.103)	1.019 (0.663)	1.032 (1.024)	1.024 (0.827)	1.026 (0.864)	1.036 (1.101)	1.029 (0.979)	1.030 (0.995)
SKILL	1.103 (0.983)	1.088 (0.885)	1.137 (1.336)	1.110 (1.095)	1.125 (1.177)	1.104 (1.024)	1.107 (1.001)	1.122 (1.158)	1.114 (1.071)	1.096 (0.959)
Observations	18860	18860	18860	18860	18860	18860	18860	18860	18860	18860

z-statistics based on robust standard errors clustered at the country level are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. All models include country, year, and industry fixed effects

Table 8 The adoption of GEPs and process

	GEP1	GEP2	GEP3	GEP4	GEP5	GEP6	GEP7	GEP8	GEP9	GEP10
GEP	1.725*** (5.521)	1.830*** (5.690)	2.794*** (11.221)	1.909*** (5.879)	2.012*** (7.285)	1.791*** (8.066)	1.632*** (5.781)	2.158*** (9.997)	1.780*** (4.453)	1.900*** (7.212)
CFL	1.578* (2.054)	1.665* (2.461)	1.514 (1.823)	1.610* (2.068)	1.616* (2.047)	1.680* (2.293)	1.637* (2.256)	1.634* (2.159)	1.572* (2.007)	1.725* (2.273)
DEBT	2.057*** (5.062)	2.091*** (5.292)	1.942*** (4.678)	2.061*** (5.277)	2.045*** (5.370)	2.074*** (5.578)	2.087*** (5.444)	2.014*** (4.958)	2.073*** (5.146)	2.088*** (5.614)
SIZE	1.130*** (3.444)	1.139*** (3.802)	1.110** (2.946)	1.120*** (3.397)	1.128*** (3.297)	1.132*** (3.366)	1.136*** (3.603)	1.118** (3.011)	1.140*** (3.629)	1.138*** (3.637)
FO	1.001 (0.879)	1.001 (0.986)	1.001 (0.677)	1.001 (0.533)	1.001 (0.653)	1.001 (0.662)	1.001 (0.690)	1.001 (1.209)	1.001 (0.856)	1.001 (0.843)
LARGE	0.998 (−1.584)	0.998 (−1.616)	0.998 (−1.752)	0.998 (−1.582)	0.998 (−1.299)	0.998 (−1.614)	0.998 (−1.640)	0.998 (−1.597)	0.998 (−1.572)	0.998 (−1.666)
AGE	0.961 (−0.847)	0.963 (−0.749)	0.971 (−0.624)	0.951 (−1.043)	0.964 (−0.759)	0.953 (−0.997)	0.958 (−0.896)	0.966 (−0.723)	0.962 (−0.831)	0.961 (−0.815)
SKILL	0.935 (−0.481)	0.917 (−0.657)	0.969 (−0.221)	0.944 (−0.420)	0.948 (−0.403)	0.933 (−0.526)	0.927 (−0.584)	0.944 (−0.442)	0.947 (−0.402)	0.919 (−0.646)
Observations	18860	18860	18860	18860	18860	18860	18860	18860	18860	18860

z-statistics based on robust standard errors clustered at the country level are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. All models include country, year, and industry fixed effects

Table 9 The GEP index and firm innovation, excluding India

Variables	Product	z-statistics	Process	z-statistics
	Odds ratio		Odds ratio	
GEP	1.185***	(10.411)	1.214***	(13.655)
CFL	1.615***	(4.572)	2.061***	(3.773)
DEBT	1.337***	(4.037)	1.728***	(6.486)
SIZE	1.089**	(2.637)	1.065	(1.798)
FO	0.999	(−1.108)	1.000	(−0.336)
LARGE	0.997**	(−2.773)	0.998	(−1.360)
AGE	1.023	(0.666)	0.950	(−0.951)
SKILL	1.155	(1.506)	0.992	(−0.055)
Observations	11374		11,217	

z-statistics based on robust standard errors clustered at the country level are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. All models include country, year, and industry fixed effects

respectively, than non-adopters. For product innovation, other practices yield odds ratios ranging from 1.515 to 1.840, indicating that adopting any single practice is associated with a 51.5% to 84.0% increase in the likelihood of innovation. Similarly, for process innovation, other practices yield odds ratios ranging from 1.632 to 2.158, representing a 63.2% to 115.8% increase in the odds of implementation.

4.2.3 Robustness checks

In this subsection, we address potential selection bias stemming from the disproportionate representation of Indian firms, which comprise 39.69% of the total sample. To ensure the results are not driven by sample composition or country-specific factors, we perform robustness checks by excluding Indian firms from the analysis. The findings remain consistent: the adoption of GEPs significantly stimulates firm innovation (Tables 9, 10 and 11).

Table 10 The adoption of GEPs and product, excluding India

	GEP1	GEP2	GEP3	GEP4	GEP5	GEP6	GEP7	GEP8	GEP9	GEP10
GEP	1.885*** (9.859)	1.565*** (5.356)	2.516*** (11.848)	1.941*** (7.671)	1.944*** (5.928)	1.535*** (5.361)	1.646*** (5.336)	1.797*** (7.537)	1.849*** (6.791)	1.577*** (5.188)
CFL	1.567*** (3.318)	1.606** (3.123)	1.482** (3.287)	1.604*** (3.916)	1.589*** (3.547)	1.632*** (3.529)	1.608*** (3.552)	1.587*** (3.591)	1.550*** (3.392)	1.661*** (3.753)
DEBT	1.406*** (4.644)	1.441*** (5.155)	1.337*** (3.957)	1.412*** (4.898)	1.413*** (5.117)	1.440*** (5.180)	1.435*** (5.140)	1.396*** (4.700)	1.412*** (4.637)	1.448*** (5.236)
SIZE	1.156*** (4.689)	1.178*** (5.488)	1.141*** (4.407)	1.150*** (4.594)	1.164*** (4.551)	1.175*** (4.871)	1.169*** (5.080)	1.163*** (4.768)	1.172*** (5.008)	1.181*** (5.281)
FO	0.999 (−0.619)	1.000 (−0.411)	0.999 (−0.727)	0.999 (−0.827)	0.999 (−0.733)	0.999 (−0.585)	0.999 (−0.703)	1.000 (−0.286)	0.999 (−0.628)	1.000 (−0.441)
LARGE	0.997** (−2.986)	0.997** (−3.015)	0.997** (−2.989)	0.997** (−2.907)	0.997* (−2.560)	0.997** (−3.028)	0.997** (−3.054)	0.997** (−2.855)	0.997** (−2.857)	0.997** (−3.070)
AGE	1.035 (0.982)	1.040 (1.227)	1.045 (1.276)	1.024 (0.739)	1.039 (1.156)	1.031 (0.966)	1.032 (0.972)	1.045 (1.243)	1.037 (1.125)	1.039 (1.193)
SKILL	1.151 (1.407)	1.122 (1.159)	1.181 (1.732)	1.152 (1.479)	1.163 (1.452)	1.135 (1.255)	1.146 (1.304)	1.165 (1.518)	1.160 (1.468)	1.118 (1.087)
Observations	11374	11374	11374	11374	11,374	11374	11374	11374	11374	11374

z-statistics based on robust standard errors clustered at the country level are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. All models include country, year, and industry fixed effects

Table 11 The adoption of GEPs and product, excluding India

	GEP1	GEP2	GEP3	GEP4	GEP5	GEP6	GEP7	GEP8	GEP9	GEP10
GEP	1.839*** (6.753)	1.821*** (5.597)	2.922*** (11.724)	2.072*** (8.021)	2.056*** (6.844)	1.784*** (6.807)	1.718*** (6.669)	2.247*** (10.589)	1.974*** (6.660)	1.953*** (6.605)
CFL	1.923*** (3.989)	1.983*** (4.111)	1.834*** (3.490)	1.962*** (3.877)	1.973*** (3.876)	2.033*** (4.118)	1.976*** (4.077)	1.978*** (3.874)	1.922*** (3.973)	2.106*** (3.995)
DEBT	1.817*** (6.691)	1.848*** (7.429)	1.715*** (6.084)	1.825*** (7.289)	1.817*** (7.465)	1.850*** (7.581)	1.853*** (7.581)	1.774*** (6.905)	1.828*** (6.950)	1.866*** (7.383)
SIZE	1.154*** (4.236)	1.166*** (4.955)	1.136*** (3.927)	1.141*** (3.981)	1.157*** (4.415)	1.160*** (4.404)	1.160*** (4.383)	1.146*** (4.064)	1.165*** (4.423)	1.163*** (4.455)
FO	1.000 (0.326)	1.000 (0.475)	1.000 (0.101)	1.000 (−0.071)	1.000 (0.085)	1.000 (0.122)	1.000 (0.112)	1.001 (0.694)	1.000 (0.287)	1.000 (0.326)
LARGE	0.998 (−1.410)	0.998 (−1.391)	0.998 (−1.558)	0.998 (−1.339)	0.998 (−1.073)	0.998 (−1.401)	0.998 (−1.410)	0.998 (−1.398)	0.998 (−1.373)	0.998 (−1.467)
AGE	0.964 (−0.716)	0.969 (−0.571)	0.977 (−0.457)	0.955 (−0.882)	0.968 (−0.605)	0.958 (−0.810)	0.963 (−0.731)	0.972 (−0.557)	0.967 (−0.663)	0.968 (−0.614)
SKILL	0.981 (−0.136)	0.951 (−0.364)	1.011 (0.078)	0.986 (−0.097)	0.983 (−0.123)	0.963 (−0.276)	0.965 (−0.270)	0.984 (−0.119)	0.994 (−0.041)	0.938 (−0.461)
Observations	11217	11217	11217	11217	11217	11217	11217	11217	11217	11217

z-statistics based on robust standard errors clustered at the country level are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. All models include country, year, and industry fixed effects

5 Conclusion and implications

This study investigates the impact of GEP adoption on firm-level product and process innovation. Using a WBES dataset of 18,860 observations, a composite GEP index was constructed using data from the Green Economy module. The logistic regression results indicate that firms that implement more green practices are more likely to engage in both product and process innovation. Further analysis of the ten individual dimensions of GEPs, including (1) heating and cooling improvements; (2) climate-friendly on-site

energy generation; (3) machinery upgrades; (4) energy management systems; (5) waste minimization, recycling, and management; (6) air pollution control; (7) water management; (8) fleet upgrades (vehicles, vessels, or aircraft); (9) lighting system improvements; and (10) other pollution control measures, reveals that each dimension is positively associated with firm innovation. Based on these results, several implications are proposed below.

5.1 Theoretical implications

This study contributes to the literature by linking environmental management and innovation from three theoretical perspectives: Institutional Theory, the Resource-Based View, and the Porter Hypothesis. First, the findings provide empirical support for Institutional Theory. Previous studies often suggest that firms may adopt green practices as symbolic actions to satisfy regulators or external stakeholders. However, our results indicate that firms respond to regulatory and social pressures through substantive organizational changes that make them more likely to engage in innovation. This result suggests that conformity to environmental expectations does not necessarily weaken competitiveness. Instead, complying with green standards can serve as a foundation for improving internal processes and developing new capabilities. In this way, the study offers a more nuanced understanding of how institutional pressures can contribute to firm-level innovation.

Second, our results extend the Resource-Based View by showing that GEPs are more than just operational costs; they are a source of valuable organizational capabilities. While the initial investment often involves physical capital, such as machinery and energy systems, the real value lies in the technical knowledge and improved routines that firms develop during the implementation process. These firm-specific strengths may be difficult for competitors to replicate. Our study confirms that these practices should be viewed as strategic resources that enhance efficiency and directly support long-term product and process innovation.

Third, the findings provide support for the strong version of the Porter Hypothesis. By examining a wide range of green practices, we show that environmental initiatives can lead to innovation offsets. Instead of being a permanent financial drain, activities like redesigning processes or upgrading technology help firms become more efficient over time. Our study demonstrates that environmental responsibility is not a burden; rather, it acts as a driver for long-term competitiveness. This shifts the focus away from a win-lose trade-off and toward a win-win scenario, where green standards encourage, rather than hinder, technological creativity.

5.2 Managerial and policy implications

5.2.1 Managerial implications

The findings of this study offer several important implications for managers seeking to enhance innovation through environmental initiatives. First, the positive association between GEP adoption and both product and process innovation suggests that green practices should not be viewed merely as regulatory compliance or cost burdens. Instead, managers can approach environmental initiatives as strategic investments that stimulate technological upgrading. Firms that systematically adopt a broader portfolio of

green practices are more likely to enhance their innovative capacity, indicating that environmental strategy can serve as a catalyst for competitive differentiation.

Second, the results show that each of the ten individual dimensions of GEPs is positively linked to innovation outcomes. This implies that managers do not need to rely on a single flagship initiative to generate innovative benefits. Operational improvements, such as heating and cooling upgrades, lighting system modernization, waste management systems, or energy management tools, can individually contribute to innovation. Therefore, managers may pursue incremental and modular green investments that fit their resource constraints while still generating innovation-enhancing effects.

5.2.2 Policy implications

The findings of this study also suggest several important implications for policymakers aiming to promote sustainable development and innovation-led growth. First, the positive relationship between GEP adoption and both product and process innovation suggests that environmental policies can yield dual benefits: environmental improvement and technological advancement. Rather than imposing excessive compliance burdens, well-designed environmental regulations may stimulate firms to modernize production systems and introduce innovative products. Policymakers should therefore view green regulations not only as environmental protection instruments but also as mechanisms to enhance national innovation capacity.

Second, because the adoption of multiple green practices appears to strengthen innovation outcomes, policymakers should design integrated policy frameworks rather than fragmented initiatives. Coordinated measures, combining environmental standards, green financing schemes, R&D subsidies, and technical advisory programs, can encourage firms to adopt complementary green practices and build long-term capabilities. Such systemic support is particularly important for small and medium-sized enterprises that may face resource constraints.

5.3 Limitations and future research

Despite the significant insights, this study is subject to several limitations that pave the way for future research. First, while the GEP index and its ten dimensions capture various aspects of green practices, they may not fully reflect the intensity, quality, or effectiveness of implementation within firms. The current measures primarily indicate whether specific practices have been adopted, but they do not capture the depth of integration or the level of managerial commitment underlying these initiatives. Future research could rely on in-depth interviews or case studies to examine how firms internally execute GEPs, thereby providing a more nuanced understanding of capability development.

Second, this study relies on cross-sectional data, which limits the ability to observe how the relationship between GEPs and innovation evolves over time. Because green investments and innovation outcomes rarely occur simultaneously, establishing a clear cause-and-effect relationship remains challenging. Future research should prioritize longitudinal designs to track the long-term development of green capabilities. Such approaches would help identify potential time-lag effects and clarify the period between the initial adoption of green practices and the realization of innovation benefits.

Third, while the cross-country evidence provides a broad overview, it may not fully capture how national institutional environments influence the relationship between GEPs and innovation. National programs, such as carbon pricing mechanisms or green R&D grants, vary significantly across countries and can act as important moderating factors. Future research should examine these country-specific policies to determine how the regulatory framework in a firm's home country strengthens or weakens the impact of GEPs on innovation outcomes. This would provide a more detailed understanding of how government support interacts with firm-level green initiatives.

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Author contributions

Quynh Trang Phan was responsible for the conceptualization, data collection, analysis, and writing of the manuscript, as well as the preparation of all figures and tables. The author reviewed and approved the final version of the manuscript.

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Data availability

The data that support the findings of this study are available at [<https://www.enterprisesurveys.org/en/data>].

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